**Lab activity – BERT**

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**BERT (Bidirectional Encoder Representations from Transformers)**

## 1. Introduction

**BERT**, which stands for **Bidirectional Encoder Representations from Transformers**, was introduced by Google researchers. It marked a massive shift in Natural Language Processing (NLP) because, unlike previous models that read text linearly (left-to-right or right-to-left), BERT reads the entire sequence of words at once.

In the context of your previous work on **Encoder-Decoder** models, BERT is an **Encoder-only** architecture. It is designed to "understand" the context of a word based on all of its surroundings (both left and right), which is why it is called "Bidirectional."

## 2. Description

The core innovation of BERT is its ability to create **contextualized word embeddings**.

- **Traditional Embeddings (Word2Vec):** The word "bank" would have the same mathematical vector whether you are talking about a "river bank" or a "savings bank."
- **BERT Embeddings:** BERT looks at the surrounding words. If it sees "water" and "flow," the vector for "bank" changes to reflect the geographical meaning. If it sees "money" and "interest," it shifts to the financial meaning.

**Key Training Strategies:**

1. **Masked Language Modeling (MLM):** BERT hides (masks) 15% of the words in a sentence and tries to predict them. This forces the model to learn deep context.
2. **Next Sentence Prediction (NSP):** BERT is trained to predict if "Sentence B" actually follows "Sentence A" in a paragraph. This helps it understand long-term relationships between sentences.

## 3. Architecture

BERT's architecture is essentially a stack of **Transformer Encoders**. Depending on the version, it uses different numbers of layers:

- **BERT-Base:** 12 layers (transformer blocks), 768 hidden units, and 12 attention heads.
- **BERT-Large:** 24 layers, 1024 hidden units, and 16 attention heads.

**The Three Embedding Layers:**

Before the text enters the encoder, it is processed through three distinct layers:

1. **Token Embeddings:** The actual words converted into vectors (using WordPiece tokenization).
2. **Segment Embeddings:** Indicators (like 0 or 1) that tell the model which sentence the word belongs to (Sentence A or Sentence B).
3. **Position Embeddings:** Since BERT reads everything at once, it doesn't naturally know the order of words. These embeddings add a "time signal" so the model knows where each word sits in the sequence.

**Comparison of BERT and GPT model:**

| **Feature** | **BERT** | **GPT (Generative Pre-trained Transformer)** |
|---|---|---|
| **Type** | Encoder-Only | Decoder-Only |
| **Direction** | Bidirectional (Looks left and right) | Unidirectional (Looks only at previous words) |
| **Primary Goal** | Understanding / Classification | Text Generation |
| **Example** | Search engines, Sentiment analysis | Chatbots, Story writing |